Solutions to Axler's Linear Algebra

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3E Products and Quotients of Vector Spaces

Exercise 3E1

Suppose T is a function from V to W. The *graph of T* is the subset of $V \times W$ defined by

$$\text{graph of } T = \{(v, Tv) \in V \times W \mid v \in V\}.$$

Prove that T is a linear map if and only if the graph of T is a subspace of $V \times W$.

Solution. Let $G(T)$ denote the graph of T. Now, suppose T is a linear map. Then $G(T)$ is nonempty since $(0, T0) = (0, 0) \in G(T)$. Now, let $(v_1, Tv_1), (v_2, Tv_2) \in G(T)$. Then

$$(v_1 - v_2, Tv_1 - Tv_2) = (v_1, Tv_1) - (v_2, Tv_2) \in G(T)$$

since T is linear. Now, let $(v, Tv) \in G(T)$ and $\alpha \in \mathbb{F}$. Then

$$(\alpha v, \alpha Tv) = \alpha(v, Tv) \in G(T)$$

since T is linear. Thus, $G(T)$ is a subspace of $V \times W$.

Conversely, suppose $G(T)$ is a subspace of $V \times W$. By definition, we have $(v_1, Tv_1), (v_2, Tv_2) \in G(T)$ for all $v_1, v_2 \in V$. Since $G(T)$ is a subspace of $V \times W$, then for all $\alpha, \beta \in \mathbb{F}$, we have

$$(\alpha v_1 + \beta v_2, \alpha Tv_1 + \beta Tv_2) = \alpha(v_1, Tv_1) + \beta(v_2, Tv_2) \in G(T).$$

By definition of $G(T)$, we have $\alpha Tv_1 + \beta Tv_2 = T(\alpha v_1 + \beta v_2)$. Thus, T is a linear map. ■

Exercise 3E2

Suppose that $V_1, \dots, V_m$ are vector spaces such that $V_1 \times \cdots \times V_m$ is finite-dimensional. Prove that V_k is finite-dimensional for each $k = 1, \dots, m$.

Solution. Let $V = V_1 \times \cdots \times V_m$, and consider $T \in \mathcal{L}(()V, V_k)$ defined by $T(v_1, \dots, v_m) = v_k$. Then T is a linear map since for all $\alpha, \beta \in \mathbb{F}$ and all $(v_1, \dots, v_m), (w_1, \dots, w_m) \in V$, we have

$$\begin{aligned} T(\alpha(v_1, \dots, v_m) + \beta(w_1, \dots, w_m)) &= T(\alpha v_1 + \beta w_1, \dots, \alpha v_m + \beta w_m) \\ &= \alpha v_k + \beta w_k \\ &= \alpha T(v_1, \dots, v_m) + \beta T(w_1, \dots, w_m). \end{aligned}$$

Since V is finite-dimensional and T is a linear map, it follows that V_k is finite-dimensional. ■

Exercise 3E3

Suppose $V_1, \dots, V_m$ are vector spaces. Prove that $\mathcal{L}(V_1 \times \cdots \times V_m, W)$ and $\mathcal{L}(V_1, W) \times \cdots \times \mathcal{L}(V_m, W)$ are isomorphic vector spaces.

Solution. Let $V = V_1 \times \cdots \times V_m$. Let $L = \mathcal{L}(V_1, W) \times \cdots \times \mathcal{L}(V_m, W)$. Define $T \in \mathcal{L}(\mathcal{L}(V, W), L)$ by

$$T(S) = (S \circ \iota_1, \dots, S \circ \iota_m)$$

where $\iota_k \in \mathcal{L}(V_k, V)$ is the map defined as

$$\iota_k(v_k) = (0, \ldots, 0, v_k, 0, \ldots, 0)$$

with $\iota_k(v_k)$ having v_k in the k-th coordinate and 0 in all other coordinates. Then T is linear since for all $\alpha, \beta \in \mathbb{F}$ and all $S_1, S_2 \in \mathcal{L}(V, W)$, we have

$$\begin{aligned} T(\alpha S_1 + \beta S_2) &= ((\alpha S_1 + \beta S_2) \circ \iota_1, \ldots, (\alpha S_1 + \beta S_2) \circ \iota_m) \\ &= (\alpha(S_1 \circ \iota_1) + \beta(S_2 \circ \iota_1), \ldots, \alpha(S_1 \circ \iota_m) + \beta(S_2 \circ \iota_m)) \\ &= \alpha T(S_1) + \beta T(S_2), \end{aligned}$$

hence T is a linear map. Now suppose $T(S) = 0$. Then $S \circ \iota_k = 0$ for all $k = 1, \ldots, m$. Moreover, for all $(v_1, \ldots, v_m) \in V$, we have

$$S(v_1, \ldots, v_m) = S(\iota_1(v_1) + \cdots + \iota_m(v_m)) = S \circ \iota_1(v_1) + \cdots + S \circ \iota_m(v_m) = 0.$$

It follows that $S(v_1, \ldots, v_m) = 0$ for all $(v_1, \ldots, v_m) \in V$. Thus, $S = 0$, and hence T is injective.

Now, let $(S_1, \ldots, S_m) \in L$. Define $S \in \mathcal{L}(V, W)$ by $S(v_1, \ldots, v_m) = S_1 v_1 + \cdots + S_m v_m$. Then $T(S) = (S_1, \ldots, S_m)$. Thus, T is surjective. Therefore, T is an isomorphism. ■

Exercise 3E4

Suppose $W_1, \ldots, W_m$ are vector spaces. Prove that $\mathcal{L}(V, W_1 \times \cdots \times W_m)$ and $\mathcal{L}(V, W_1) \times \cdots \times \mathcal{L}(V, W_m)$ are isomorphic vector spaces.

Remark. *Note the difference between this exercise and Exercise 3E3. Here, the domain is decomposed, while in Exercise 3E3, the codomain is decomposed. Hence, the use of the maps ι_k and π_k in the solutions to these exercises are reversed.*

Solution. Let $W = W_1 \times \cdots \times W_m$. Let $L = \mathcal{L}(V, W_1) \times \cdots \times \mathcal{L}(V, W_m)$. Define $T \in \mathcal{L}(\mathcal{L}(V, W), L)$ by

$$T(S) = (\pi_1 \circ S, \ldots, \pi_m \circ S)$$

where $\pi_k \in \mathcal{L}(W, W_k)$ is the map defined as

$$\pi_k(w_1, \ldots, w_m) = w_k$$

with $\pi_k(w_1, \ldots, w_m)$ having w_k in the k-th coordinate. Then T is linear since for all $\alpha, \beta \in \mathbb{F}$ and all $S_1, S_2 \in \mathcal{L}(V, W)$, we have

$$\begin{aligned} T(\alpha S_1 + \beta S_2) &= (\pi_1 \circ (\alpha S_1 + \beta S_2), \ldots, \pi_m \circ (\alpha S_1 + \beta S_2)) \\ &= (\alpha(\pi_1 \circ S_1) + \beta(\pi_1 \circ S_2), \ldots, \alpha(\pi_m \circ S_1) + \beta(\pi_m \circ S_2)) \\ &= \alpha T(S_1) + \beta T(S_2), \end{aligned}$$

hence T is a linear map. Now suppose $T(S) = 0$. Then $\pi_k \circ S = 0$ for all $k = 1, \ldots, m$. Moreover, for all $v \in V$, we have

$$Sv = (\pi_1(Sv), \ldots, \pi_m(Sv)) = (0, 0, 0) = 0.$$

It follows that $Sv = 0$ for all v, hence $S = 0$, and thus T is injective.

Now let $(S_1, \ldots, S_m) \in L$. Define $S \in \mathcal{L}(V, W)$ by $Sv = (S_1 v, \ldots, S_m v)$. Then $T(S) = (S_1, \ldots, S_m)$. Thus, T is surjective. Therefore, T is an isomorphism. ■

Exercise 3E5

For m a positive integer, define V^m by

$$V^m = \underbrace{V \times \cdots \times V}_{m \text{ times}}.$$

Prove that V^m and $\mathcal{L}(\mathbb{F}^m, V)$ are isomorphic vector spaces.

Solution. Let $e_1, \ldots, e_m$ be the standard basis of $\mathbb{F}^m$. Define $T \in \mathcal{L}(V^m, \mathcal{L}(\mathbb{F}^m, V))$ by

$$T(v_1, \ldots, v_m)(\lambda_1, \ldots, \lambda_m) = \lambda_1 v_1 + \cdots + \lambda_m v_m.$$

Then T is linear since for all $\alpha, \beta \in \mathbb{F}$ and all $v = (v_1, \ldots, v_m), w = (w_1, \ldots, w_m) \in V^m$, we have

$$\begin{aligned} T(\alpha v + \beta w)(\lambda_1, \ldots, \lambda_m) &= T(\alpha v_1 + \beta w_1, \ldots, \alpha v_m + \beta w_m)(\lambda_1, \ldots, \lambda_m) \\ &= \lambda_1(\alpha v_1 + \beta w_1) + \cdots + \lambda_m(\alpha v_m + \beta w_m) \\ &= \alpha(\lambda_1 v_1 + \cdots + \lambda_m v_m) + \beta(\lambda_1 w_1 + \cdots + \lambda_m w_m) \\ &= \alpha T(v)(\lambda_1, \ldots, \lambda_m) + \beta T(w)(\lambda_1, \ldots, \lambda_m), \end{aligned}$$

hence T is a linear map. Now suppose $T(v) = 0$. Then $T(v)(e_k) = 0$ for all $k = 1, \ldots, m$. It follows that $v_k = 0$ for all $k = 1, \ldots, m$, hence $v = 0$, and thus T is injective.

Now let $S \in \mathcal{L}(\mathbb{F}^m, V)$. Define $v \in V^m$ by $v_k = Se_k$ for all $k = 1, \ldots, m$. Then $T(v) = S$. Thus, T is surjective. Therefore, T is an isomorphism. ■

Exercise 3E6

Suppose that v, x are vectors in V and that U, W are subspaces of V such that $v + U = x + W$. Prove that $U = W$.

Solution. Let $u \in U$. Then $v + u \in v + U = x + W$. Thus, there exists $w \in W$ such that $v + u = x + w$. It follows that $u = (x - v) + w \in W$, hence $U \subseteq W$. Now let $w \in W$. Then $x + w \in x + W = v + U$. Thus, there exists $u \in U$ such that $x + w = v + u$. It follows that $w = (v - x) + u \in U$, hence $W \subseteq U$. Therefore, $U = W$. ■

Exercise 3E7

Let $U = \{(x, y, z) \in \mathbb{R}^3 \mid 2x + 3y + 5z = 0\}$. Suppose $A \subseteq \mathbb{R}^3$. Prove that A is a translate of U if and only if there exists $c \in \mathbb{R}$ such that

$$A = \{(x, y, z) \in \mathbb{R}^3 \mid 2x + 3y + 5z = c\}.$$

Solution. Suppose A is a translate of U. Then there exists $v \in \mathbb{R}^3$ such that $A = v + U$. Let

$v = (x_0, y_0, z_0)$. Then

$$\begin{aligned} A &= v + U \\ &= \{(x_0, y_0, z_0) + (x, y, z) \in \mathbb{R}^3 \mid 2x + 3y + 5z = 0\} \\ &= \{(x', y', z') \in \mathbb{R}^3 \mid 2(x' - x_0) + 3(y' - y_0) + 5(z' - z_0) = 0\} \\ &= \{(x', y', z') \in \mathbb{R}^3 \mid 2x' + 3y' + 5z' = 2x_0 + 3y_0 + 5z_0\}. \end{aligned}$$

Conversely, suppose there exists $c \in \mathbb{R}$ such that

$$A = \{(x, y, z) \in \mathbb{R}^3 \mid 2x + 3y + 5z = c\}.$$

Let $v = (x_0, y_0, z_0)$ where x_0, y_0, z_0 are any real numbers such that $2x_0 + 3y_0 + 5z_0 = c$. Then

$$\begin{aligned} v + U &= (x_0, y_0, z_0) + U \\ &= \{(x_0, y_0, z_0) + (x, y, z) \in \mathbb{R}^3 \mid 2x + 3y + 5z = 0\} \\ &= A. \end{aligned}$$

■

Exercise 3E8

(a) Suppose $T \in \mathcal{L}(V, W)$ and $c \in W$. Prove that $\{x \in V \mid Tx = c\}$ is either the empty set or is a translate of $\operatorname{null} T$.

(b) Explain why the set of solutions to a system of linear equations such as 3.27 is either the empty set or is a translate of some subspace of $\mathbb{F}^n$.

Solution.

(a) Suppose $T \in \mathcal{L}(V, W)$ and $c \in W$. Let $A = \{x \in V \mid Tx = c\}$. If A is nonempty, then there exists $v \in V$ such that $Tv = c$. We claim that $A = v + \operatorname{null} T$. Let $x \in A$. Then $Tx = c = Tv$, hence $T(x - v) = 0$, and thus $x - v \in \operatorname{null} T$. It follows that $x \in v + \operatorname{null} T$, hence $A \subseteq v + \operatorname{null} T$. Now let $y \in v + \operatorname{null} T$. Then there exists $u \in \operatorname{null} T$ such that $y = v + u$. It follows that $Ty = T(v + u) = Tv + Tu = c + 0 = c$, hence $y \in A$, and thus $v + \operatorname{null} T \subseteq A$. Therefore, $A = v + \operatorname{null} T$.

(b) A system of linear equations such as 3.27 can be written in the form $Tx = c$ for some linear map T and some vector c. By part (a), the set of solutions to this system of linear equations is either the empty set or is a translate of $\operatorname{null} T$, which is a subspace of $\mathbb{F}^n$. ■

Exercise 3E9

Prove that a nonempty subset A of V is a translate of some subspace of V if and only if $\lambda v + (1 - \lambda)w \in A$ for all $v, w \in A$ and all $\lambda \in \mathbb{F}$.

Solution. Suppose A is a translate of some subspace of V. Then there exists $v \in V$ and a subspace U of V such that $A = v + U$. Let $w \in A$. Then there exists $u \in U$ such that $w = v + u$. Now let $\lambda \in \mathbb{F}$. Then

$$\lambda v + (1 - \lambda)w = \lambda v + (1 - \lambda)(v + u) = v + (1 - \lambda)u \in A.$$

Conversely, suppose $\lambda v + (1-\lambda)w \in A$ for all $v, w \in A$ and all $\lambda \in \mathbb{F}$. Fix some $v_0 \in A$, and let $U = \{u \in V \mid v_0 + u \in A\}$. We first claim that U is a subspace of V. Let $u_1, u_2 \in U$. Then $v_0 + u_1, v_0 + u_2 \in A$. Now let $\lambda \in \mathbb{F}$. Then

$$\begin{aligned}\lambda(v_0 + u_1) + (1-\lambda)(v_0 + u_2) &= v_0 + \lambda u_1 + (1-\lambda)u_2 \\ &= v_0 + (\lambda u_1 + (1-\lambda)u_2) \in A.\end{aligned}$$

It follows that $\lambda u_1 + (1-\lambda)u_2 \in U$, hence U is a subspace of V. Now we claim that $A = v_0 + U$. Let $w \in A$. By definition, $v_0 + (w - v_0) = w \in A$, hence $w - v_0 \in U$, and thus $w \in v_0 + U$, so $A \subseteq v_0 + U$. Now let $v_0 + u \in v_0 + U$. Then $v_0 + u \in A$ by definition of U, hence $v_0 + U \subseteq A$. Therefore, $A = v_0 + U$. ■

Exercise 3E10

Suppose $A_1 = v + U_1$ and $A_2 = w + U_2$ for some $v, w \in V$ and some subspaces U_1, U_2 of V. Prove that the intersection $A_1 \cap A_2$ is either a translate of some subspace of V or is the empty set.

Solution. If $A_1 \cap A_2$ is the empty set, then we are done. Suppose that $A_1 \cap A_2$ is nonempty. Let $x, y \in A_1 \cap A_2$ with $\lambda \in \mathbb{F}$. Then $x, y \in A_1$, hence $\lambda x + (1-\lambda)y \in A_1$. Moreover, $x, y \in A_2$, hence $\lambda x + (1-\lambda)y \in A_2$. It follows that $\lambda x + (1-\lambda)y \in A_1 \cap A_2$. Thus, by Exercise 3E9, $A_1 \cap A_2$ is a translate of some subspace of V. ■

Exercise 3E11

Suppose $U = \{(x_1, x_2, \ldots) \in \mathbb{F}^\infty \mid x_k \neq 0$ for only finitely many $k\}$.

(a) Show that U is a subspace of $\mathbb{F}^\infty$.

(b) Prove that $\mathbb{F}^\infty / U$ is infinite-dimensional.

Solution.

(a) Since $(0, \ldots) \in U$, then U is nonempty. Let $u = (u_1, u_2, \ldots), v = (v_1, v_2, \ldots) \in U$ and $\lambda \in \mathbb{F}$. Then $u_k \neq 0$ for only finitely many k, and $v_k \neq 0$ for only finitely many k. It follows that $\lambda u_k + (1-\lambda)v_k \neq 0$ for only finitely many k, hence $\lambda u + (1-\lambda)v \in U$. Thus, U is a subspace of $\mathbb{F}^\infty$.

(b) For any given positive integer m, define the sets B_i for $1 \leq i \leq m$ as follows:

$$B_i = \{i + km \mid k \in \mathbb{N}\} = \{i, i+m, i+2m, \ldots\}.$$

Then $B_1, \ldots, B_m$ are pairwise disjoint subsets of $\mathbb{N}$. For each $i = 1, \ldots, m$, define the j-th coordinate of $v_i \in \mathbb{F}^\infty$ by

$$v_{i,j} = \begin{cases} 1 & \text{if } j \in B_i \\ 0 & \text{otherwise} \end{cases}$$

Hence, $v_1 = (1, 0, \ldots, 0, 1, 0, \ldots)$ with 1's in the 1-st, $(m+1)$-st, $(2m+1)$-st, … coordinates and $v_2 = (0, 1, 0, \ldots)$ with 1's in the 2-nd, $(m+2)$-nd, $(2m+2)$-nd, … coordinates and so on. We claim that $v_1 + U, \ldots, v_m + U$ are linearly independent in $\mathbb{F}^\infty / U$. Let $\lambda_1, \ldots, \lambda_m \in \mathbb{F}$ such that $\lambda_1(v_1 + U) + \cdots + \lambda_m(v_m + U) = U$. Then $\lambda_1 v_1 + \cdots + \lambda_m v_m \in U$. Then each λ_i for

$i = 1, \dots, m$ is 0 as follows. For each i, consider the j-th coordinate of v_i for $j \in B_i$. Since v_i has 1's in all coordinates indexed by B_i and 0's elsewhere, and B_i is disjoint from B_k for $k \neq i$, the j-th coordinate of $\lambda_1 v_1 + \cdots + \lambda_m v_m$ is λ_i. But this sum is in U, which means it has only finitely many nonzero coordinates. Since B_i is infinite, it follows that $\lambda_i = 0$. Hence, all $\lambda_i = 0$, proving that $v_1 + U, \dots, v_m + U$ are linearly independent. Since m was arbitrary, $\mathbb{F}^\infty / U$ is infinite-dimensional. ■

Exercise 3E12

Suppose $v_1, \dots, v_m \in V$. Let

$$A = \{\lambda_1 v_1 + \cdots + \lambda_m v_m \mid \lambda_1, \dots, \lambda_m \in \mathbb{F} \text{ and } \lambda_1 + \cdots + \lambda_m = 1\}.$$

(a) Prove that A is a translate of some subspace of V.

(b) Prove that if B is a translate of some subspace of V and $\{v_1, \dots, v_m\} \subseteq B$, then $A \subseteq B$.

(c) Prove that A is a translate of some subspace of V of dimension less than m.

Solution.

(a) Consider any element of A, say $\lambda_1 v_1 + \cdots + \lambda_m v_m$ with $\lambda_1 + \cdots + \lambda_m = 1$. Since the λ_i's sum to 1, a natural choice for scalars are

$$\lambda_i = \mu_i + \frac{1}{m}, \quad i = 1, \dots, m, \text{ with } \mu_1 + \cdots + \mu_m = 0.$$

We claim that the set of all elements

$$A' = \{\mu_1 v_1 + \cdots + \mu_m v_m \mid \mu_1 + \cdots + \mu_m = 0\}$$

form a subspace of V. To see this, note that $0 \in A'$ since we can take $\mu_1 = \cdots = \mu_m = 0$. Moreover, if $x, y \in A'$, say $x = \mu_1 v_1 + \cdots + \mu_m v_m$ and $y = \nu_1 v_1 + \cdots + \nu_m v_m$ with $\mu_1 + \cdots + \mu_m = 0$ and $\nu_1 + \cdots + \nu_m = 0$, then for any scalar $\alpha \in \mathbb{F}$, we have

$$\alpha x + y = (\alpha\mu_1 + \nu_1) v_1 + \cdots + (\alpha\mu_m + \nu_m) v_m,$$

and $\alpha\mu_1 + \nu_1 + \cdots + \alpha\mu_m + \nu_m = \alpha(\mu_1 + \cdots + \mu_m) + (\nu_1 + \cdots + \nu_m) = 0$. Hence, $\alpha x + y \in A'$, proving that A' is indeed a subspace of V.

Finally, note that any element of A can be written as

$$\lambda_1 v_1 + \cdots + \lambda_m v_m = \left(\frac{1}{m}(v_1 + \cdots + v_m)\right) + (\mu_1 v_1 + \cdots + \mu_m v_m).$$

This shows that $A \subseteq \frac{1}{m}(v_1 + \cdots + v_m) + A'$. To see the reverse inclusion, note that any element of $\frac{1}{m}(v_1 + \cdots + v_m) + A'$ is of the form

$$\frac{1}{m}(v_1 + \cdots + v_m) + (\mu_1 v_1 + \cdots + \mu_m v_m) = (\mu_1 + \frac{1}{m}) v_1 + \cdots + (\mu_m + \frac{1}{m}) v_m,$$

with $\mu_1 + \cdots + \mu_m = 0$. Setting $\lambda_i = \mu_i + \frac{1}{m}$, we have $\lambda_1 + \cdots + \lambda_m = 1$, and hence this element belongs to A. Therefore, $A = \frac{1}{m}(v_1 + \cdots + v_m) + A'$, proving that A is indeed a translate of the subspace A'.

(b) Suppose B is a translate of some subspace of V, say $B = v + U$ for some $v \in V$ and subspace $U \subseteq V$, and $\{v_1, \dots, v_m\} \subseteq B$. Then for each i, we can write $v_i = v + u_i$ with $u_i \in U$. Consider any element of A, say $\lambda_1 v_1 + \cdots + \lambda_m v_m$ with $\lambda_1 + \cdots + \lambda_m = 1$. We have

$$\begin{aligned}\lambda_1 v_1 + \cdots + \lambda_m v_m &= \lambda_1(v + u_1) + \cdots + \lambda_m(v + u_m)\\ &= (\lambda_1 + \cdots + \lambda_m)v + (\lambda_1 u_1 + \cdots + \lambda_m u_m)\\ &= v + (\lambda_1 u_1 + \cdots + \lambda_m u_m).\end{aligned}$$

Since U is a subspace, $\lambda_1 u_1 + \cdots + \lambda_m u_m \in U$, and hence $\lambda_1 v_1 + \cdots + \lambda_m v_m \in v + U = B$. Therefore, $A \subseteq B$.

(c) To see that $\dim A' < m$, observe that $\mu_1 + \cdots + \mu_m = 0$ imposes a linear constraint on the μ_i's. In particular, observe that

$$\mu_1 = -(\mu_2 + \cdots + \mu_m).$$

Hence, any element of A' can be written as

$$\mu_1 v_1 + \cdots + \mu_m v_m = -(\mu_2 + \cdots + \mu_m)v_1 + \mu_2 v_2 + \cdots + \mu_m v_m = \mu_2(v_2 - v_1) + \cdots + \mu_m(v_m - v_1),$$

which shows that A' is spanned by $v_2 - v_1, \dots, v_m - v_1$. Therefore, $\dim A' \leq m - 1 < m$. ■

Exercise 3E13

Suppose U is a subspace of V such that V/U is finite-dimensional. Prove that V is isomorphic to $U \times (V/U)$.

Solution. Since V/U is finite-dimensional, let $v_1 + U, \dots, v_m + U$ be a basis of V/U. Let $W = \operatorname{span}(v_1, \dots, v_m)$. We show that $V = U \oplus W$. First, note that any $v \in V$ can be written as $v = u + w$ for some $u \in U$ and $w \in W$. Indeed, since $v + U \in V/U$, we can write

$$v + U = \lambda_1(v_1 + U) + \cdots + \lambda_m(v_m + U)$$

for some scalars $\lambda_1, \dots, \lambda_m$. This gives

$$v - (\lambda_1 v_1 + \cdots + \lambda_m v_m) \in U.$$

So setting $u = v - (\lambda_1 v_1 + \cdots + \lambda_m v_m) \in U$ and $w = \lambda_1 v_1 + \cdots + \lambda_m v_m \in W$, we have $v = u + w$. To see that the sum is direct, suppose $u + w = 0$ with $u \in U$ and $w \in W$. Then $w = -u \in W \cap U$. But if $w = \lambda_1 v_1 + \cdots + \lambda_m v_m \in U$, then $w + U = 0 + U$ in V/U, which implies $\lambda_1 = \cdots = \lambda_m = 0$ since $v_1 + U, \dots, v_m + U$ are linearly independent in V/U. Hence, $w = 0$ and $u = 0$, proving that $V = U \oplus W$.

Consider the map $T : V \to U \times V/U$ given by $Tv = (u, v + U)$, where $v = u + w$ with $u \in U$ and $w \in W$. This map is well-defined because the decomposition $v = u + w$ is unique. It is linear, injective, and surjective, hence an isomorphism. Therefore, $V \cong U \times V/U$. ■

Exercise 3E14

Suppose U and W are subspaces of V and $V = U \oplus W$. Suppose $w_1, \dots, w_m$ is a basis of W. Prove that $w_1 + U, \dots, w_m + U$ is a basis of V/U.

Solution. Since $V = U \oplus W$, every $v \in V$ can be written uniquely as $v = u + w$ with $u \in U$ and $w \in W$. We claim that $w_1 + U, \ldots, w_m + U$ is a basis of V/U. First, we show that they span V/U. Let $v + U \in V/U$. Write $v = u + w$ with $u \in U$ and $w \in W$. Then $v + U = (u + w) + U = w + U$. Since $w \in W$, we can write $w = \lambda_1 w_1 + \cdots + \lambda_m w_m$ for some scalars $\lambda_1, \ldots, \lambda_m$. Hence, $v + U = \lambda_1(w_1 + U) + \cdots + \lambda_m(w_m + U)$, showing that $w_1 + U, \ldots, w_m + U$ span V/U.

Next, we show linear independence. Suppose $\lambda_1(w_1 + U) + \cdots + \lambda_m(w_m + U) = 0 + U$. Then $\lambda_1 w_1 + \cdots + \lambda_m w_m \in U$. But $\lambda_1 w_1 + \cdots + \lambda_m w_m \in W$ and $U \cap W = \{0\}$, so $\lambda_1 w_1 + \cdots + \lambda_m w_m = 0$. Since $w_1, \ldots, w_m$ are linearly independent, we have $\lambda_1 = \cdots = \lambda_m = 0$. Therefore, $w_1 + U, \ldots, w_m + U$ are linearly independent, and hence form a basis of V/U. ■

Exercise 3E15

Suppose U is a subspace of V and $v_1 + U, \ldots, v_m + U$ is a basis of V/U and $u_1, \ldots, u_n$ is a basis of U. Prove that $v_1, \ldots, v_m, u_1, \ldots, u_n$ is a basis of V.

Solution. First, we show that $v_1, \ldots, v_m, u_1, \ldots, u_n$ span V. Let $v \in V$. Then $v + U \in V/U$, so we can write

$$v + U = \lambda_1(v_1 + U) + \cdots + \lambda_m(v_m + U)$$

for some scalars $\lambda_1, \ldots, \lambda_m$. This gives $v - (\lambda_1 v_1 + \cdots + \lambda_m v_m) \in U$. Since $u_1, \ldots, u_n$ is a basis of U, we can write $v - (\lambda_1 v_1 + \cdots + \lambda_m v_m) = \mu_1 u_1 + \cdots + \mu_n u_n$ for some scalars $\mu_1, \ldots, \mu_n$. Hence, $v = \lambda_1 v_1 + \cdots + \lambda_m v_m + \mu_1 u_1 + \cdots + \mu_n u_n$, showing that $v_1, \ldots, v_m, u_1, \ldots, u_n$ span V.

Next, we show linear independence. Suppose $\lambda_1 v_1 + \cdots + \lambda_m v_m + \mu_1 u_1 + \cdots + \mu_n u_n = 0$. Then $\lambda_1 v_1 + \cdots + \lambda_m v_m \in U$. Then

$$\lambda_1(v_1 + U) + \cdots + \lambda_m(v_m + U) = 0 + U.$$

Since $v_1 + U, \ldots, v_m + U$ are linearly independent in V/U, we must have $\lambda_1 = \cdots = \lambda_m = 0$. Then the original equation reduces to $\mu_1 u_1 + \cdots + \mu_n u_n = 0$, and since $u_1, \ldots, u_n$ are linearly independent, we have $\mu_1 = \cdots = \mu_n = 0$. Therefore, $v_1, \ldots, v_m, u_1, \ldots, u_n$ are linearly independent, and hence form a basis of V. ■

Exercise 3E16

Suppose $\varphi \in \mathcal{L}(V, \mathbb{F})$ and $\varphi \neq 0$. Prove that $\dim V/(\operatorname{null} \varphi) = 1$.

Solution. From 3.107 in the text, we know that $V/\operatorname{null} \varphi \cong \mathbb{F}$. Since $\dim \mathbb{F} = 1$, it follows that $\dim V/(\operatorname{null} \varphi) = 1$. ■

Exercise 3E17

Suppose U is a subspace of V such that $\dim V/U = 1$. Prove that there exists $\varphi \in \mathcal{L}(V, \mathbb{F})$ such that $\operatorname{null} \varphi = U$.

Solution. Since $\dim V/U = \dim \mathbb{F} = 1$, then there exists some isomorphism $\psi \in \mathcal{L}(V/U, \mathbb{F})$. Let $\pi : V \to V/U$ be the quotient map. Define $\varphi = \psi \circ \pi \in \mathcal{L}(V, \mathbb{F})$. Suppose $\varphi(v) = 0$. Then $\psi(\pi(v)) = 0$, which implies $\pi(v) = 0$ since ψ is an isomorphism. Hence, $v \in U$, and we have $\operatorname{null}\varphi \subseteq U$. Conversely, if $v \in U$, then $\pi(v) = 0$, so $\varphi(v) = \psi(\pi(v)) = 0$. Therefore, $\operatorname{null}\varphi = U$. ■

Exercise 3E18

Suppose that U is a subspace of V such that V/U is finite-dimensional.

(a) Show that if W is a finite-dimensional subspace of V and $V = U + W$, then $\dim W \geq \dim V/U$.

(b) Prove that there exists a finite-dimensional subspace W of V such that $\dim W = \dim V/U$ and $V = U \oplus W$.

Solution.

(a) Suppose W is a finite-dimensional subspace of V and $V = U + W$. Let $\pi : V \to V/U$ be the quotient map, and consider the restriction $\pi|_W : W \to V/U$. To see that $\operatorname{range}(\pi|_W) = V/U$, note that for any $v + U \in V/U$, we can write $v = u + w$ for some $u \in U$ and $w \in W$. Then $\pi(v) = \pi(u + w) = \pi(u) + \pi(w) = 0 + \pi(w) = \pi(w)$, so $v + U = \pi(w) \in \operatorname{range}(\pi|_W)$. Hence, $\operatorname{range}(\pi|_W) = V/U$. From the rank-nullity theorem, we have

$$\dim W = \dim \operatorname{null} \pi|_W + \dim \operatorname{range}(\pi|_W) = \dim \operatorname{null} \pi|_W + \dim V/U \geq \dim V/U.$$

(b) Let $v_1 + U, \ldots, v_m + U$ be a basis of V/U. We claim that $v_1, \ldots, v_m$ are linearly independent. Suppose $\lambda_1 v_1 + \cdots + \lambda_m v_m = 0$ for some scalars $\lambda_1, \ldots, \lambda_m$. Then

$$\lambda_1 v_1 + \cdots + \lambda_m v_m \in U.$$

But $\lambda_1(v_1 + U) + \cdots + \lambda_m(v_m + U) = 0$ in V/U, and since $v_1 + U, \ldots, v_m + U$ are linearly independent, it follows that $\lambda_1 = \cdots = \lambda_m = 0$. Hence, $v_1, \ldots, v_m$ are linearly independent.

Let $W = \operatorname{span}\{v_1, \ldots, v_m\}$. Then $\dim W = m = \dim V/U$. For any $v \in V$, we can write

$$v + U = \lambda_1(v_1 + U) + \cdots + \lambda_m(v_m + U).$$

This implies that

$$v - (\lambda_1 v_1 + \cdots + \lambda_m v_m) \in U,$$

so $v \in U + W$. Hence, $V = U + W$. To see that $U \cap W = \{0\}$, let $w \in U \cap W$. Then $w \in W$, so $w = \lambda_1 v_1 + \cdots + \lambda_m v_m$ for some scalars $\lambda_1, \ldots, \lambda_m$. But $w \in U$, so $w + U = 0$ in V/U. Hence,

$$\lambda_1(v_1 + U) + \cdots + \lambda_m(v_m + U) = 0$$

in V/U. Since $v_1 + U, \ldots, v_m + U$ are linearly independent, it follows that $\lambda_1 = \cdots = \lambda_m = 0$. Hence, $w = 0$, and we have $U \cap W = \{0\}$. Therefore, $V = U \oplus W$. ■

Exercise 3E19

Suppose $T \in \mathcal{L}(V, W)$ and U is a subspace of V. Let π denote the quotient map from V onto V/U. Prove that there exists $S \in \mathcal{L}(V/U, W)$ such that $T = S \circ \pi$ if and only if $U \subseteq \operatorname{null} T$.

Solution. Suppose $S \in \mathcal{L}(V/U, W)$ is such that $T = S \circ \pi$. Then for any $u \in U$, we have $T(u) = S(\pi(u)) = S(0) = 0$, so $U \subseteq \operatorname{null} T$.

Conversely, suppose $U \subseteq \operatorname{null} T$. Define $S : V/U \to W$ by $S(v + U) = T(v)$. This is well-defined because if $v + U = v' + U$, then $v - v' \in U \subseteq \operatorname{null} T$, so $T(v) = T(v')$. It is straightforward to check that S is linear and that $T = S \circ \pi$. ■